

Software Requirements Specification

1. Functional Requirements

FR-1 User Authentication & Role Management

- The system shall allow users to log in as Gamer, Audience, Community Manager, System Moderator, or System Administrator.
- The system shall authorize actions based on the user's role.
- Unauthorized users shall be denied access to restricted features.

FR-2: Community and Channel Management

- The system shall allow a Gamer to create and publish a game community after validating community details.
- The system shall assign the creator as the Community Owner and make published communities visible to other users.
- The system shall allow the Community Owner to create multiple channels within a community.
- The system shall allow configuration of channel types and permissions.
- The system shall store and manage community and channel details in the database.

FR-3 Discover and Join Community

- The system shall allow Audience users to browse available communities.
- The system shall display community details and rules.
- The system shall allow Audience users to join a community.
- The system shall store community membership information.

FR-4 Channel Interaction and Engagement

- The system shall allow users to post messages and react to content in channels.
- The system shall update the activity feed in real time.
- The system shall award XP for valid user interactions.
- The system shall store messages and reactions in the database.

FR-5 Event Monitoring and Rule Enforcement

- The system shall allow the Community Manager to monitor live events.
- The system shall detect rule violations during events.
- The system shall allow warning, muting, or penalizing participants.
- The system shall update participant status in the database.

FR-6 Reporting User or Content

- The system shall allow Gamer and Audience users to report content or users.
- The system shall allow selecting report reasons and adding descriptions.
- The system shall generate a unique report ID.
- The system shall store reports in the database.
- The system shall notify the System Moderator.

FR-7 Report Review and Action

- The system shall allow the System Moderator to review reports.
- The system shall allow validating or closing reports.
- The system shall allow actions such as warn, mute, or temporary ban.
- The system shall notify the reporting user and the reported user.
- The system shall update reports and user status in the database.

FR-8 Feedback and Analytics

- The system shall collect feedback after events.
- The system shall store feedback in the database.
- The system shall allow the System Administrator to view platform metrics.
- The system shall aggregate analytics from stored data.

FR-9 Platform Administration

- The system shall allow the System Administrator to configure moderation rules.
- The system shall allow updating automated moderation settings.
- The system shall maintain audit logs for moderation actions.

2. Non-Functional Requirements

NFR-1 Performance

- The system shall respond to user actions within acceptable time limits.
- The system shall handle multiple users interacting simultaneously.

NFR-2 Scalability

- The system shall support growth in users, communities, channels, and events.
- The system architecture shall allow future feature expansion.

NFR-3 Security

- The system shall enforce role-based access control.
- The system shall protect user data and reports from unauthorized access.
- The system shall securely store authentication and activity data.

NFR-4 Reliability

- The system shall ensure data consistency across all operations.
- The system shall prevent data loss during failures.
- The system shall maintain logs for recovery and auditing.

NFR-5 Usability

- The system shall provide an intuitive and user-friendly interface.
- The system shall clearly guide users through community, channel, and event workflows.

NFR-6 Maintainability

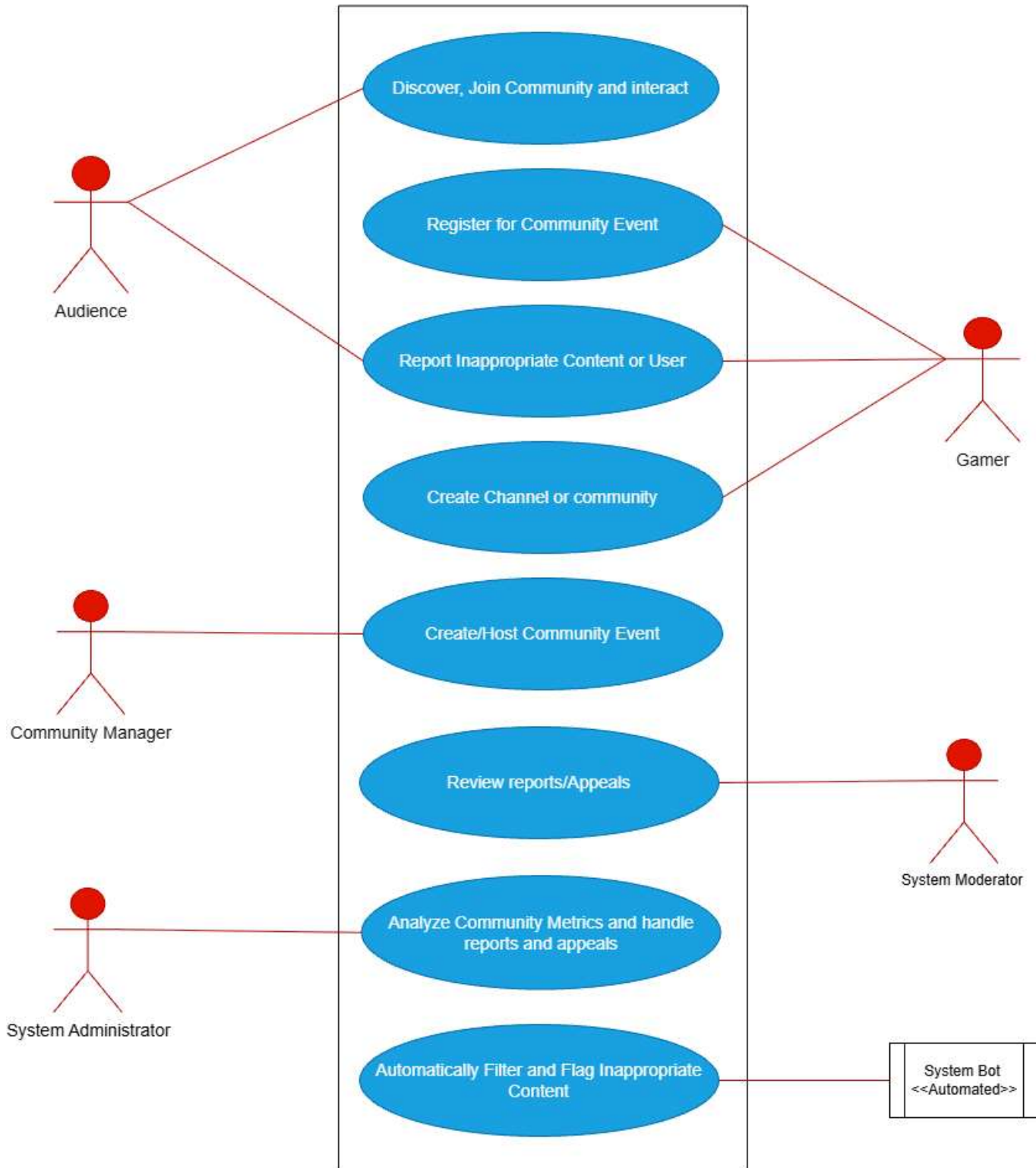
- The system shall follow MVC architecture.
- The system shall use modular and readable code.
- The system shall allow easy debugging and updates.

NFR-7 Compatibility

- The system shall run on modern web browsers.
- The system shall be accessible on different screen sizes.

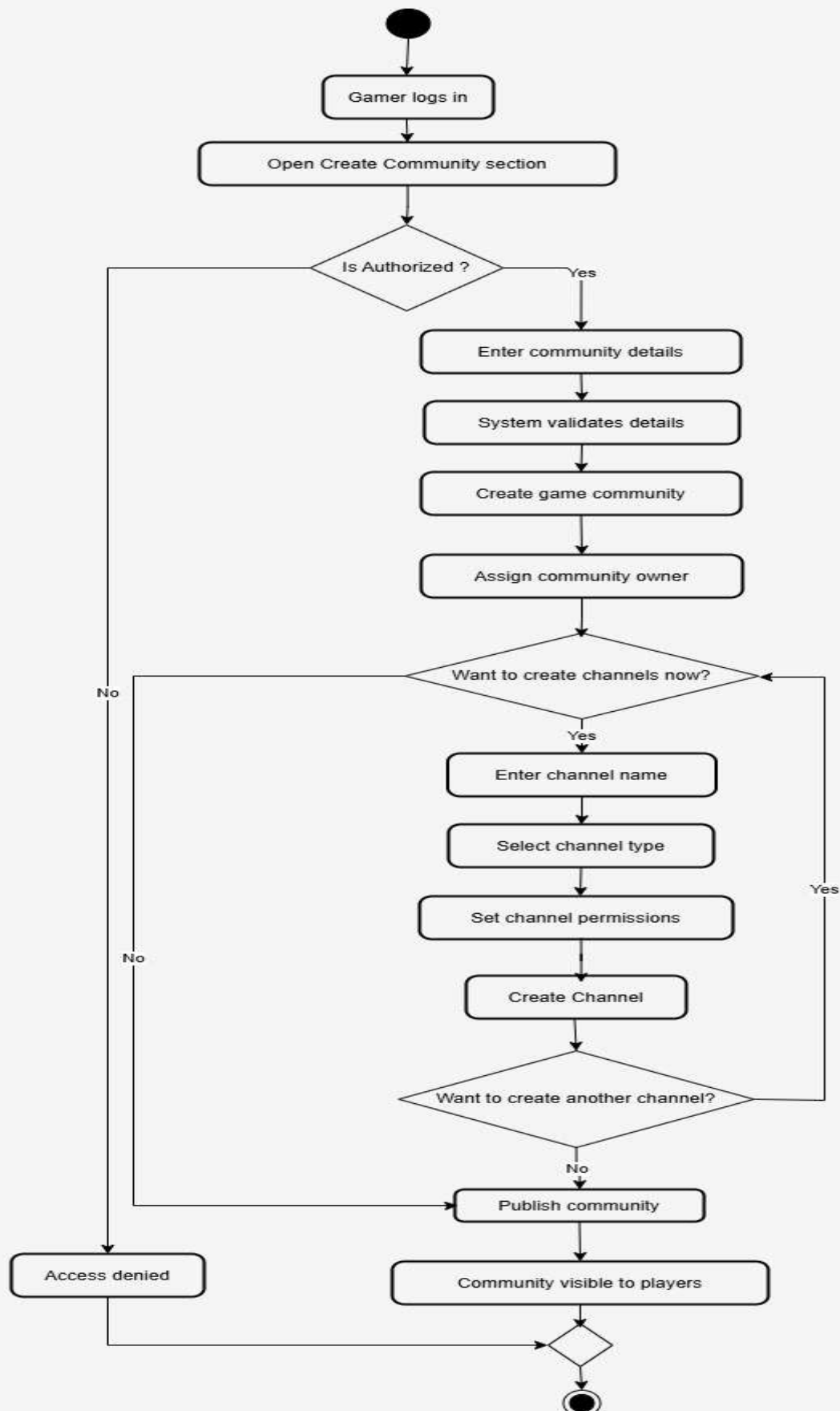
UML Diagrams

1. Use Case Diagram

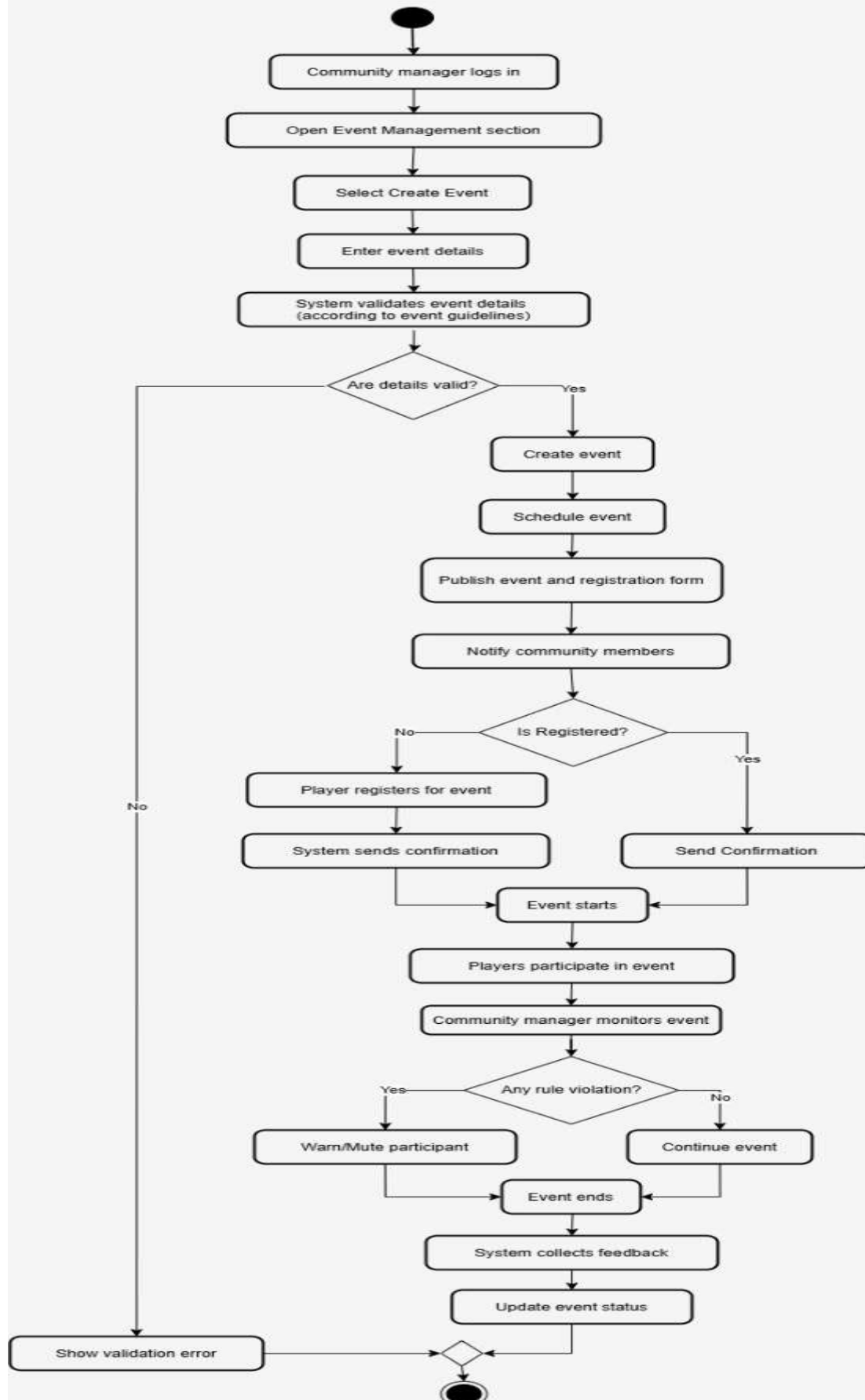


2. Activity Diagrams

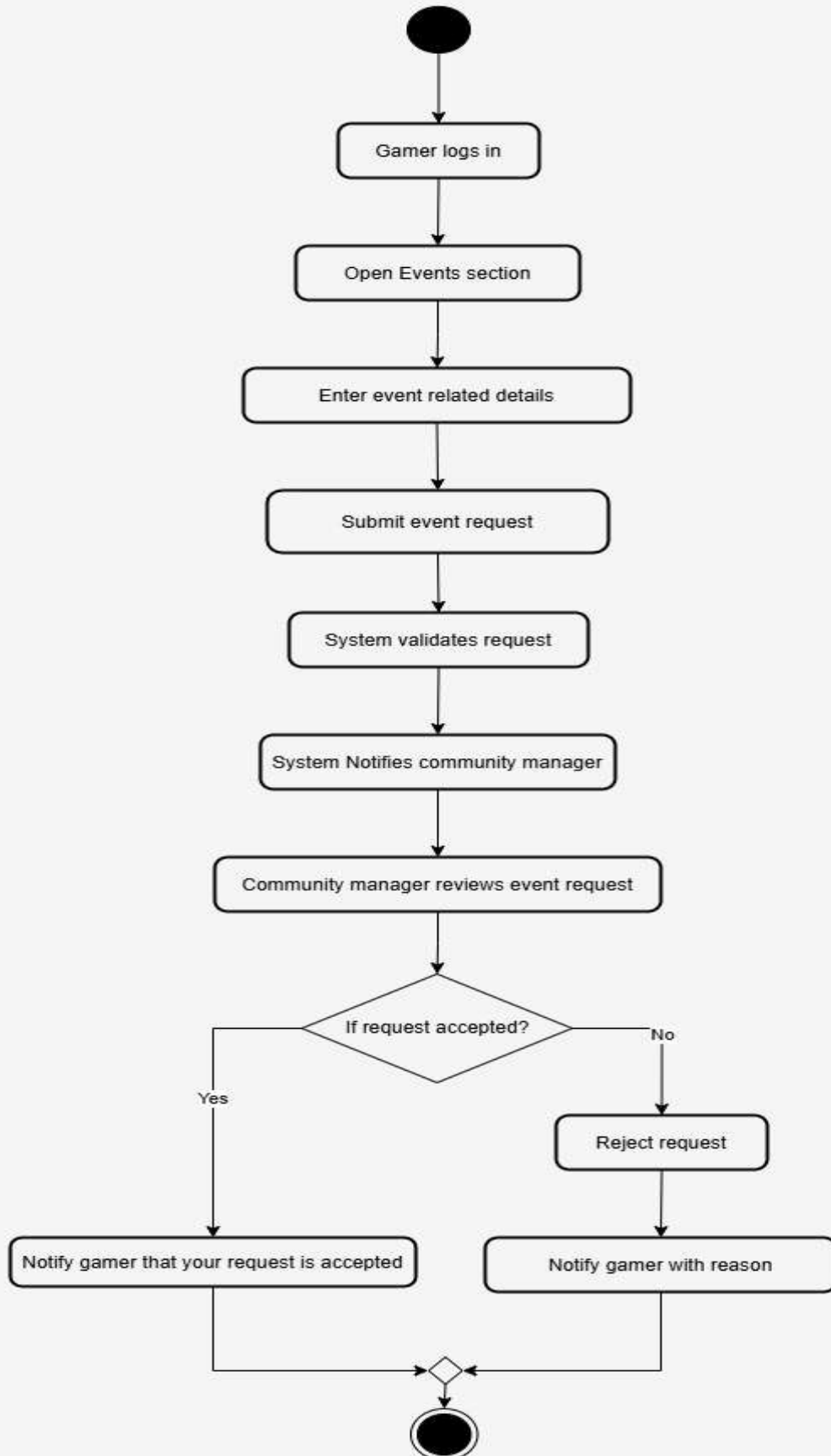
Activity Diagram: Create Game Community & Channel



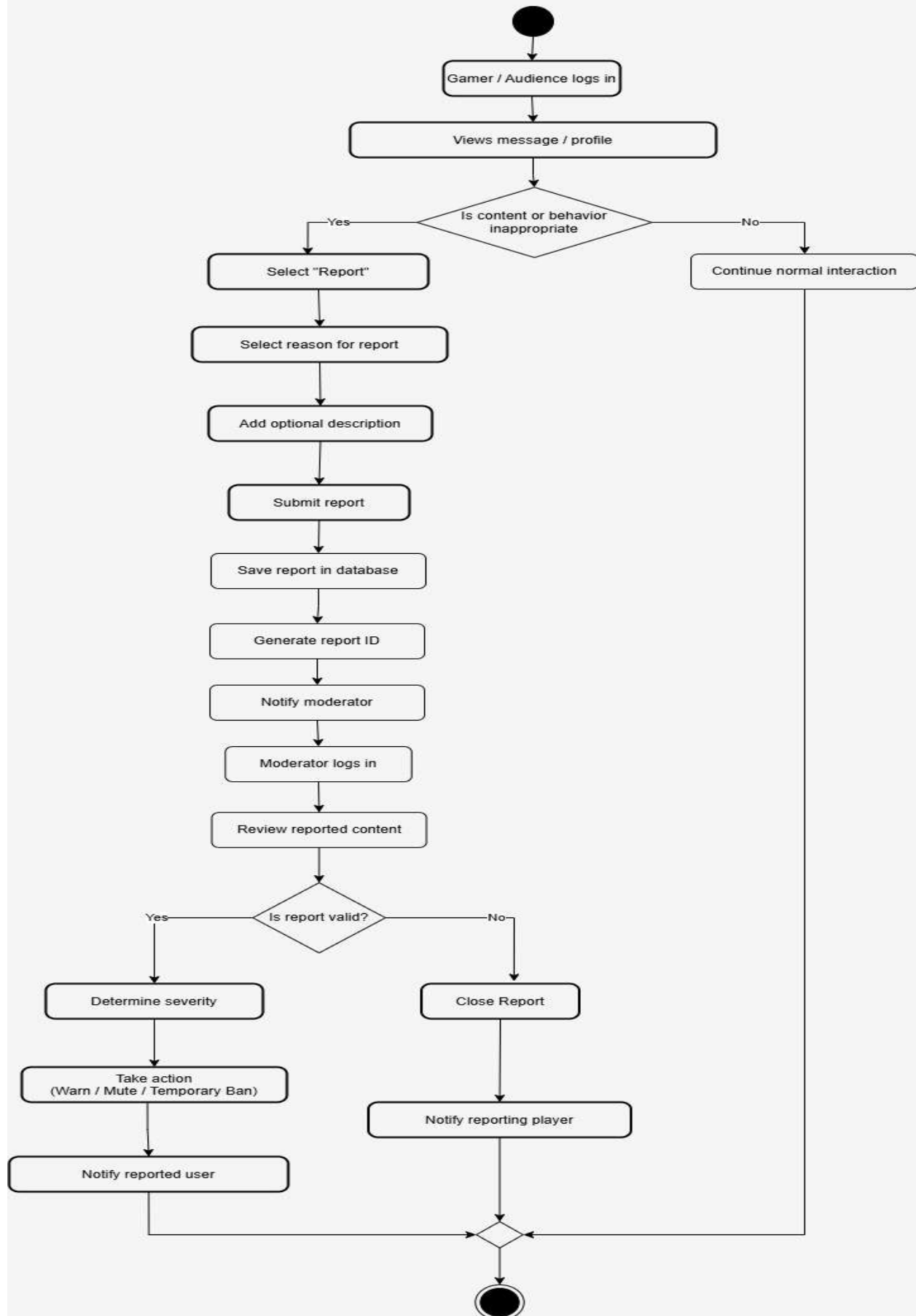
Activity Diagram: Create Community Event and member participation



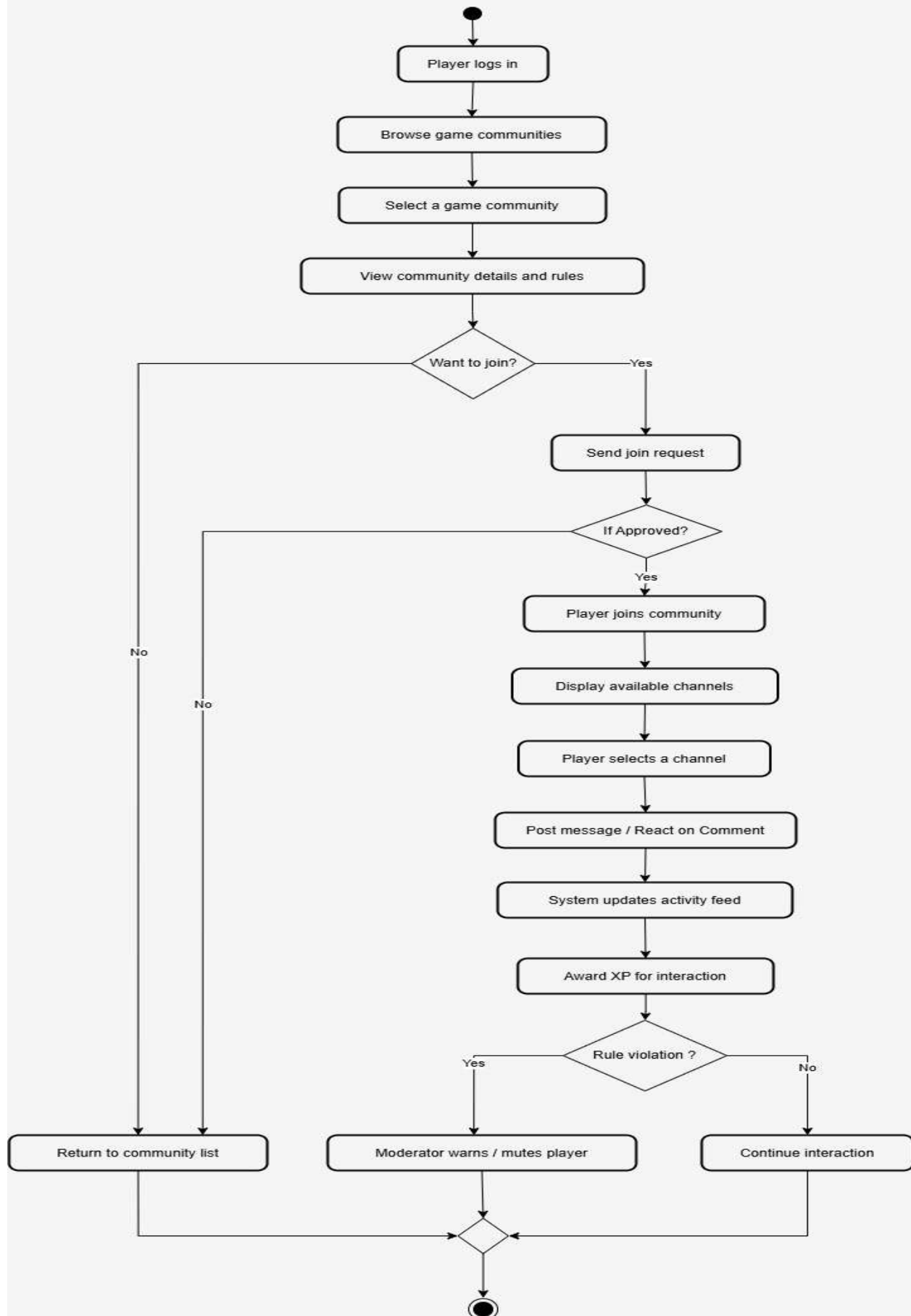
Activity Diagram: Event Request & Approval



Activity Diagram - Report User or Content and taking action



Activity Diagram: Join Game Community & Interact



3. Sequence Diagram

